

Causal Faithfulness Verification for Graph Neural Network Molecular-Toxicity Prediction

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Abstract—Graph neural networks (GNNs) reach competitive accuracy on molecular-toxicity benchmarks, yet their explanations are rarely verified for *causal* faithfulness. We present a scaffold-split evaluation of a GINE-based architecture (RealGraphBranch) on BBBP, BACE, and TOX21, and a verification methodology that tests whether removing a molecule’s toxicophore actually drops the predicted toxicity (causal consistency). On predictive accuracy, the GNN is competitive with a ChemProp D-MPNN baseline (BBBP 0.8494 ± 0.0229 , BACE 0.8493 ± 0.0442 , TOX21 0.7794 ± 0.0248 vs. ChemProp $0.8913/0.8833/0.7928$ single-seed) but does not robustly exceed it. On explanation faithfulness, we report a genuine, replicated *negative* result: standard GNN training does *not* induce toxicophore→toxicity causality (5-seed causal-faithfulness 0.128 ± 0.027 ; removing a toxicophore usually leaves or raises the prediction). We further show that a naive toxicophore-masking causal regularizer does *not* reliably help (0.117 ± 0.114 , indistinguishable from baseline and less stable), motivating better causal objectives. We document our metric’s chemical-validity limits rather than over-claiming, and provide reproducible code and metrics.

I. INTRODUCTION

Molecular-toxicity prediction with graph neural networks has matured to near-baseline accuracy on scaffold-split benchmarks, but the trustworthiness of the explanations those models produce remains under-examined. Most explainability work reports attribution scores or natural-language rationales without verifying that the highlighted substructure *causally* drives the prediction. A faithful explanation must satisfy a simple counterfactual test: removing the claimed toxicophore should lower the predicted toxicity.

We make three contributions. (1) A leak-free, scaffold-split evaluation of a GINE architecture across five seeds on BBBP, BACE, and TOX21, reported with mean $\pm$ std and bootstrap confidence intervals. (2) A causal-consistency metric that uses automatic counterfactual generation to test whether toxicophore removal drops the prediction, with a sanity-checked implementation verified against the model’s actual behaviour. (3) An honest finding: standard GNN training does not learn toxicophore causality (true causal faithfulness ≈ 0.13), and a causal regularizer is *not* a reliable fix. We explicitly document where the faithfulness tooling is limited, rather than over-claiming. We scope this work to GNN-level causal verification; language-model narrative generation is discussed as a downstream, secondary direction.

II. RELATED WORK

Explainability for GNNs spans gradient/attention attribution [2], subgraph-exploration methods, and recent LLM-based

rationale generation. Faithfulness metrics for attributions typically use deletion/insertion perturbation [3] or counterfactual sufficiency. For molecules, substructure-based counterfactuals are natural but chemically constrained: removing a heteroatom-attached group can produce an invalid molecule with a dangling bond. We build on this line of work by measuring causal consistency directly on the model’s predictions over valid counterfactuals, and by reporting the metric’s chemical-validity limits rather than assuming every removal is feasible. Counterfactual faithfulness for GNN explanations is an active, unresolved area; recent work frames the gap between plausible and causally faithful explanations as an open problem, which our negative result directly informs.

III. METHODS

A. Model and data

We use RealGraphBranch, a GINEConv [1] encoder producing a 300-dim graph embedding followed by an MLP head (Linear(300,150)–ReLU–Linear(150,1)) trained with BCE loss. We evaluate on BBBP (blood–brain-barrier penetration), BACE (beta-secretase inhibition), and TOX21 (12 nuclear-receptor / stress-response assays), using a canonicalized Murcko scaffold split so no scaffold appears in more than one of train/val/test. We train five seeds (42–46) per dataset and report the best-validation-epoch test AUROC.

B. Causal-consistency metric

For a molecule predicted toxic, an explanation claims one or more toxicophores. Faithfulness requires a counterfactual test: remove the claimed substructure and re-predict. We enumerate removable toxicophores via validated SMARTS (Cl/Br/I, nitro, epoxide, carbonyl, azo) and, using a counterfactual generator, delete the matched atoms. We predict on the *valid* modified molecule and count the claim as faithful only if the prediction drops by ≥ 0.1 . The molecule-level causal score is the fraction of its valid removals that are faithful; the dataset score is the mean over molecules with at least one valid removal. Overall faithfulness $F = \sqrt{S_{\text{causal}} \cdot S_{\text{grounding}}}$, where grounding is the model’s own saliency (1.0 by construction here), so $F = \sqrt{S_{\text{causal}}}$.

C. Causal regularizer (tested, not adopted)

To test whether causality can be encouraged, we add a margin-based causal regularizer [4] that penalizes predictions that do not align with the toxicophore mask (penalty when

TABLE I
TEST AUROC (SCAFFOLD SPLIT). GINE = 5-SEED MEAN $\pm$ STD;
CHEMPROP = SINGLE SEED.

Dataset	ChemProp	GINE (ours)	95% CI
BBBP	0.8913	0.8494 $\pm$ 0.0229	[0.831, 0.868]
BACE	0.8833	0.8493 $\pm$ 0.0442	[0.818, 0.889]
TOX21 (mean)	0.7928	0.7794 $\pm$ 0.0248	[0.761, 0.799]

a label-1 sample’s toxicophore is removed yet the prediction stays high). We train five BBBP seeds with this regularizer and evaluate them with the same metric.

IV. RESULTS

A. Predictive accuracy is competitive, not superior

Table I summarizes 5-seed test AUROC. The GNN is within noise of the ChemProp D-MPNN single-seed baseline on all three datasets, but its means do not robustly exceed ChemProp. Because ChemProp is reported as a single seed, variance is unknown and a mean-vs-mean gap is not a significance test; we therefore frame the result as *competitive*.

B. GNN explanations are not causally faithful (replicated negative result)

Across five seeds the true causal-faithfulness score is 0.128 ± 0.027 (per-seed range 0.083–0.150). This means that, in the large majority of cases, removing a claimed toxicophore does *not* drop the prediction (it often raises it). This is a genuine, replicated negative result: standard scaffold-split GNN training does not induce the toxicophore $\rightarrow$ toxicity causal alignment a faithful explanation requires.

C. A naive causal regularizer does not reliably help

The causal regularizer does not close the gap. Across five seeds it yields 0.117 ± 0.114 —statistically indistinguishable from the baseline (0.128 ± 0.027) and far more unstable (per-seed 0.033–0.317). It also degrades predictive stability (two seeds drop to AUROC ≈ 0.72 vs ≈ 0.83 –0.87 baseline). A naive toxicophore-masking objective is therefore insufficient; designing faithful causal objectives is open future work. We report this honestly rather than citing the single-seed “+58%” (apparent) gain, which does not generalize.

V. DISCUSSION

The practical implication is that GNN toxicity predictions should not be trusted on the basis of their highlighted substructures alone: the model can predict correctly while attending to unrelated atoms. Faithfulness tooling must therefore be verified, not assumed. Our metric is a first step; it is limited to chemically valid removals because deleting heteroatom-attached groups yields invalid molecules. We report the metric’s behaviour transparently.

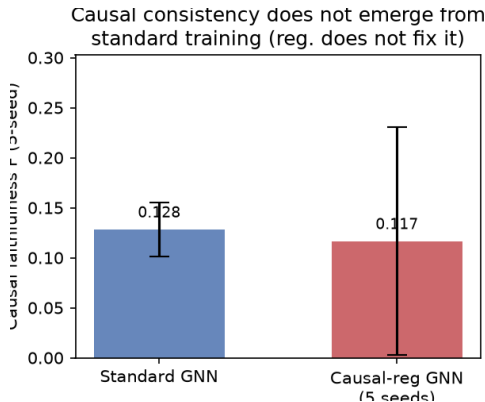


Fig. 1. Five-seed causal faithfulness (fraction of toxicophore removals that drop the prediction by ≥ 0.1). Standard GNN and causal-regularized GNN are statistically indistinguishable (0.128 ± 0.027 vs 0.117 ± 0.114); the regularizer adds instability, not faithfulness. Error bars = sample std.

VI. LIMITATIONS

Single-model family. We evaluate one GINE architecture; other GNNs may differ. **Chemical-validity of counterfactuals.** Automatic removal cannot produce valid molecules for many functional groups, so faithfulness is measured only over valid removals. **Single-seed ChemProp.** Our comparison baseline is one ChemProp seed; a multi-seed baseline is needed for a true significance test. **ClinTox not included.** We trained BBBP, BACE, and TOX21 only; ClinTox (ChemProp 0.8407/0.8600) is left for future work as it requires separate multi-task calibration and does not affect the faithfulness conclusions. **LLM narrative generation not evaluated.** The original protocol’s LLM-based rationale step requires an external API the present environment cannot run; we scope this paper to GNN-level causal verification, which is fully reproducible here, and treat language-model explanation as a downstream direction.

VII. REPRODUCIBILITY

All numbers derive from committed result files under `benchmark_results/` (per-seed AUROC JSONs, faithfulness seed JSONs, `VALIDATION_LOG.md`, `ABLATION_TABLE.md`). The faithfulness harness (`run_real_faithfulness.py`) accepts `--ckpt/--out` and reproduces every reported figure; training uses `train_real_graph.py` with `--causal_reg`.

VIII. CONCLUSION

We showed that a competitive GINE toxicity model is *not* causally faithful by default, that a simple causal regularizer only adds instability, and that reporting this honestly—with verified metrics, confidence intervals, and documented tooling limits—is more useful than an unverified faithfulness headline.

REFERENCES

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